```
import pandas as pd
import numpy as np
from sklearn import linear_model
import matplotlib.pyplot as plt
from sklearn.metrics import mean_absolute_error,r2_score
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

```
df = pd.read_csv('Housing-selected-columns.csv')
df.head()
```

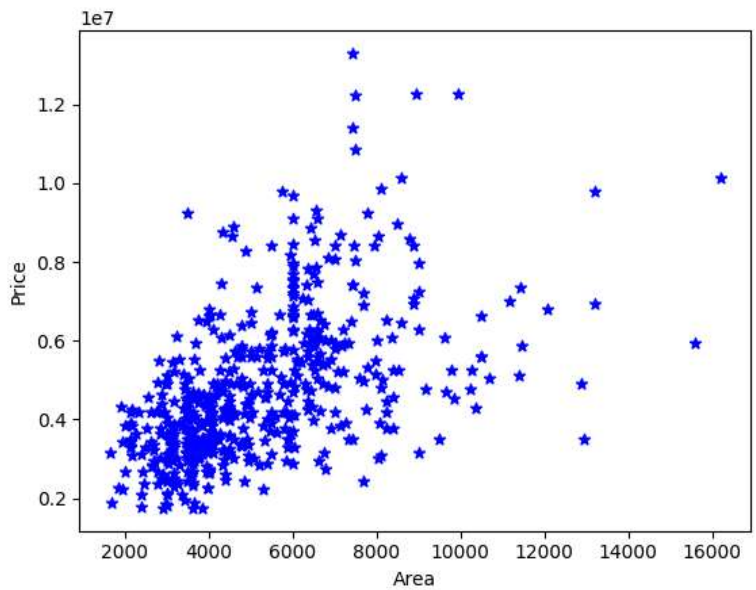
	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning
0	13300000	7420	4	2	3	yes	no	no	no	yes
1	12250000	8960	4	4	4	yes	no	no	no	yes
2	12250000	9960	3	2	2	yes	no	yes	no	no
3	12215000	7500	4	2	2	yes	no	yes	no	yes
4	11410000	7420	4	1	2	yes	yes	yes	no	yes

Next steps: Generate code with df New interactive sheet

```
%matplotlib inline

X = df[['area','bedrooms','bathrooms']]
y = df['price']
plt.scatter(df.area, df.price, color="blue", marker="*")
plt.xlabel('Area')
plt.ylabel('Price')
```

```
Text(0, 0.5, 'Price')
```




```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)
```

```
model = LinearRegression()

model.fit(X_train,y_train)
```

```
LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test)
```

```
print("R2 Score:",r2_score(y_test,y_pred))
print("MAE:",mean_absolute_error(y_test,y_pred))
```

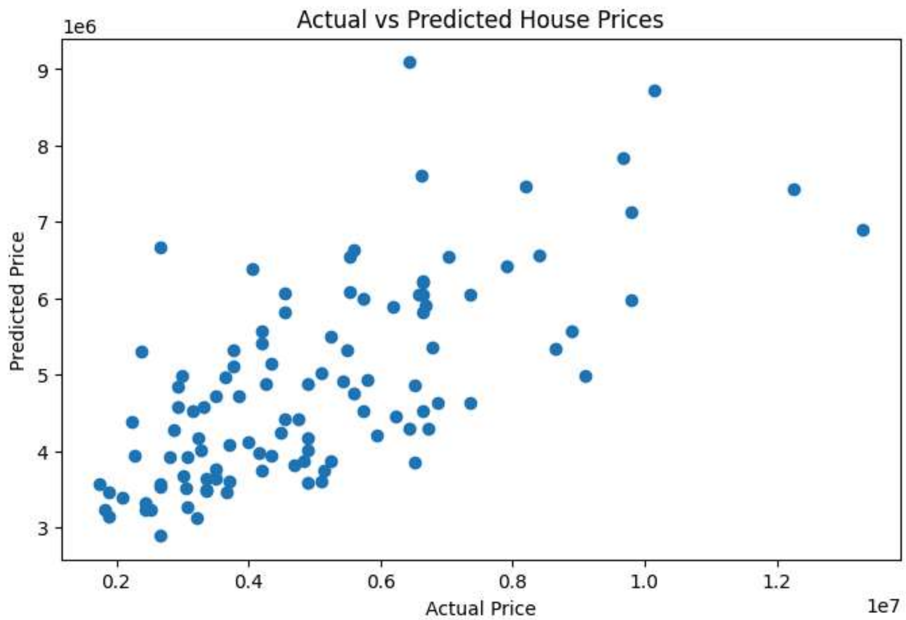
```
R2 Score: 0.4559299118872445
MAE: 1265275.6699454375
```

```
new_house = pd.DataFrame({
    'area':[2000],
    'bedrooms':[3],
    'bathrooms':[2]
})

price = model.predict(new_house)
print(price)
```

```
[4675650.78603028]
```

```
plt.figure(figsize=(8,5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")
plt.show()
```




```
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
```

```
Intercept: 59485.379208717495
Coefficients: [3.45466570e+02 3.60197650e+05 1.42231966e+06]
```

```
df.to_csv('Housing-selected-columns.csv', index=False)
```

```
from google.colab import files
files.download('Housing-selected-columns.csv')
```